

# Boya Ashok Kumar

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## Objective

Motivated B.Tech CSE-AI graduate (2026 Batch, CGPA: 8.0) targeting Infosys **Specialist Programmer** and **Digital Specialist Engineer** roles. Proficient in Python, Java, JavaScript, AI/ML, and full-stack web development. Proven track record in building production-grade intelligent systems across internships and projects.

## Technical Skills

**Languages:** Java, Python, JavaScript, SQL  
**AI / ML:** Scikit-learn, OpenCV, Pandas, NumPy  
**Web Development:** HTML5, CSS3, JavaScript, Flask, REST APIs, Streamlit  
**Databases:** MySQL, JDBC, Sevlets  
**Tools & Cloud:** Git, GitHub, AWS (EC2, S3), OOP

## Internship Experience

**Java Full Stack Intern** Jan 2026 – Present  
*Tap Academy Bengaluru, India*

- Built responsive web interfaces using **HTML, CSS, and JavaScript** following the complete **SDLC** lifecycle — from requirement gathering through testing and delivery.
- Performed **MySQL** database operations via **JDBC** and applied **OOP design patterns** (MVC, Factory) to write clean, maintainable backend code.
- Designed and executed functional test cases per **STLC** practices; logged and tracked defects systematically with written status reports for sprint reviews.

**AI / ML Intern** Jun 2025 – Aug 2025  
*Infosys Springboard Remote*

- Built a **real-time crowd density estimation system** using **PyTorch and OpenCV** with preprocessing techniques (frame extraction, normalization) to improve model accuracy.
- Developed interactive monitoring dashboards using **Streamlit and Python** to visualize density metrics and surface actionable insights for operations teams.
- Wrote structured **unit tests** for the inference pipeline following **STLC principles**; tracked all changes on **GitHub** with commit-level documentation.

## Projects

**Exposys Mart — Entropy-Based DDoS Detection System** Python, FastAPI, MySQL, Chart.js, JavaScript

- Built a **real-time intrusion detection system** using **FastAPI and Shannon Entropy algorithms** to detect volumetric DDoS attacks by identifying anomalies in network traffic patterns.
- Implemented a **persistent IP blocking mechanism** with custom middleware and **MySQL** backend, automating attacker identification via 90th-percentile traffic analysis.
- Built a **real-time analytics dashboard** with Chart.js for live visualization of entropy trends, attack severity, and traffic telemetry.

**EduMetric — Student Performance Prediction System** Flask, PostgreSQL, Python, Scikit-learn, REST APIs

- Built a full-stack analytics platform using **Flask and PostgreSQL** with modular **REST API** architecture following **OOP design patterns** across the complete SDLC lifecycle.
- Developed **predictive ML models** (Scikit-learn) to identify at-risk students early, powering an interactive dashboard with actionable academic insights for administrators.

## Deep-Vision Crowd Monitoring System

PyTorch, OpenCV, Streamlit, Python, GitHub

- Developed a **real-time crowd density and anomaly detection pipeline** using **PyTorch and OpenCV** with frame-level preprocessing for accurate, low-latency inference.
- Applied model optimization (**normalization, hyperparameter tuning**) to improve detection precision; documented the full system on GitHub with a comprehensive project report.
- Built a **live Streamlit dashboard** providing real-time density metrics, anomaly alerts, and trend visualization for proactive crowd management.

## Certifications

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**Oracle** — AI Vector Search Professional

**IBM SkillsBuild** — AI & Data Fundamentals

## Education

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**St. Johns College of Engineering and Technology**

Sep 2022 – May 2026

*B.Tech in Computer Science & Artificial Intelligence (CSE-AI)* — CGPA: **8.3 / 10**

**Deeksha Junior College**

Aug 2020 – Sep 2022

*Intermediate — MPC* — CGPA: 7.78 / 10